FORUM:	United Nations Office for Outer Space Affairs
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Introduction

Figure 1: military satellites orbiting Earth

Space, initially regarded as a realm for scientific exploration and international cooperation for technological development, is undergoing a subsequent transformation into a new security domain characterized by an intensifying strategic arms race among major state powers. As state governments increasingly rely on satellites for military communications, reconnaissance, missile detection and warning, rapid development of space technologies often culminates in accelerating competition between member states. Given that scientific development of space technologies closely aligns with state military strategies, novel space technologies to strengthen national security are often perceived as offensive threats to other nations, fueling a space arms race driven by mutual distrust and military expansion. The continuous cycle of military buildup not only triggers geopolitical conflicts among states but increases risks of inflicting widespread damage on both established military systems and associated civilian services—such as communication, navigation, finance, weather forecasts, and disaster responses. Therefore, the international community must consider establishing transnational preventative norms and frameworks, as multilateral cooperation to mitigate the arms race of space technologies and ensure peaceful use of space has become necessary more than ever.

Background

The roots of the modern arms race in outer space trace back to periods of the Cold War. Following the Soviet Union's launch of Sputnik 1, the world's first artificial satellite, in 1957, the United States and the Soviet Union started to investigate the field of space technology, marking the start of the so-called Cold War Space Race. To counter the Soviet Union's early achievements, the United States invested a massive amount of resources and finance into space exploration, spearheaded by the National Aeronautics and Space Administration (NASA). Culminating in the United States's successful Apollo 11 moon landing in 1969, which cost approximately 25.4 billion USD, the arms race demonstrated itself not only as a remarkable scientific achievement but also as large-scale strategic geopolitical competition aimed at securing national prestige and technological superiority. Furthermore, rocket technologies utilized for launching satellites are closely associated with ballistic missile technologies carrying nuclear weapons, which highlighted expanded satellite capabilities for reconnaissance, military communication, missile warnings and navigational applications during armed conflicts, in addition to fundamental civilian infrastructural services.

Aiming to impede such an arms race, the international community—including the United States and the Soviet Union—concluded the Outer Space Treaty in 1967, prohibiting the deployment of nuclear weapons and weapons associated with mass destruction in orbit in order to establish the principle of the peaceful use of outer space. Nonetheless, given that the treaty did not prohibit conventional anti-satellite weapons or the operation of military satellites, the militarization of space persistently expanded its role, even after the conclusion of the Cold War. These days, an accelerating number of states such as China and India attempt to develop anti-satellite capabilities and advanced technologies, such as signal jamming and cyberattacks, for military purposes. While the development of defensive technologies and investment in securing space technologies are considered essential for strengthening national security, an intensifying cross-state arms race in space technologies indicates growing risks of mutual distrust and arms buildups, eventually threatening international security and the safety of satellite services.

Figure 2: Representatives signing the Outer Space Treaty

Problems Raised

The security dilemma created by defensive space capabilities

The profound concern regarding an arms race of space technologies lies in the perception that defensive space capabilities, developed to strengthen state security, are often considered an offensive threat to other state governments. A known example of the issue is the Strategic Defense Initiative (SDI), announced by the United States in 1983. As the United States sought to protect itself from nuclear attacks by researching a defense mechanism of intercepting ballistic missiles using space and ground-based technologies, SDI envisioned a multi-layered defense system designed to detect and track heat signatures generated during missile launches using space-based infrared sensors. Particularly, the concept of Brilliant Pebbles, the deployment of numerous small, autonomous interceptor satellites into orbit, was examined, highlighting its attempt at direct utilization of space technologies in national security. In response, the Soviet Union expressed its concern that the system would weaken its retaliatory capabilities and geopolitical balance, arguing that it could lead to the development of space weapons and a new arms race. Consequently, the Soviet Union also sought to strengthen its response capabilities, marking a security dilemma that makes disarmament and diplomatic cooperation difficult.

The security dilemma of space technology is exemplified by the debate surrounding the deployment of THAAD in South Korea in 2016. Despite the South Korean government's prescription of THAAD as a defensive system designed to counter North Korea's missile threats, the Chinese state government expressed concern that the system's radar and missile defense capabilities could undermine its own strategic security. Consequently, the diplomatic relations between the two states deteriorated, followed by the Chinese government's measures of unofficial economic and cultural sanctions on South Korean cultural content and the tourism industry. The case of THAAD demonstrates that the defensive capabilities of a nation could escalate beyond military tension into intricate diplomatic and economic conflicts. Thus, a lack of trust regarding its intentions and the potential future application of space technologies could trigger a military arms race and diplomatic friction.

Long-term threats to peaceful space activities and the orbital environment

The space arms race threatens not only the military tensions between competing states but also the peaceful space activities and orbital environment. Considering that modern satellites are utilized for civil services including GPS, weather forecasting, financial transactions, and internet connectivity as well as military communications and reconnaissance, it is often challenging to clearly mark the distinction between military and civilian satellites. As a result, satellite attacks or radio-frequency and cyber interference carried out under state military operations often disrupt civilian services and expose dual-use satellites to attack. Furthermore, several developing nations' reliance on satellite services of their neighboring states suggests increasing competition between major powers impeding multiple states' civil services and infrastructure indirectly.

Figure 3: Dual-use satellite civilian military services

Additionally, if states conduct destructive tests aimed at securing a strategic advantage or demonstrating their anti-satellite capabilities, there could exist long-term damage to the orbital environment, extending beyond competing circumstances. In particular, the direct-ascent anti-satellite

(ASAT) missiles, which carries potent of physically destroy satellites, could induce transnational concerns associated with the long-lasting orbital environment. For instance, in 2007, China intercepted its weather satellite, Fengyun IC, using a ground-launched missile; the test generated over 3300 trackable fragments and accelerated the total number of tracked space objects by 25%, approximately. Given that the destruction occurred at an altitude of 862km, a significant amount of debris remained in orbit for an extended period, continuously posing a collision risk to satellites operating nearby. This generated space debris threatens Earth observations, scientific investigation and telecommunications; it is not merely a temporary security concern but a threat that could compromise long-term safety and accessibility of the orbital environment.

International Actions

Prevention of an Arms Race in Outer Space (PAROS)

The Outer Space Treaty of 1967 prohibited the deployment of nuclear weapons and weapons of mass destruction in Earth's orbit in order to establish a principle of peaceful use of celestial bodies; yet, the treaty does not explicitly prohibit emerging counter-space capabilities, such as ground-launched anti-satellite missiles, jamming and cyberattacks. Thus, in order to address the legal gap, the United Nations General Assembly has consistently adopted resolutions on the Prevention of an Armed Race in Outer Space (PAROS) since 1981, urging the Conference of Disarmament (CD) to formulate additional international norms. Subsequently, CD established a special committee on PAROS in 1985, aiming to address the issue of the deployment of space weapons, anti-satellite capabilities and measures for verification.

Figure 4: UN Conference on Disarmament Geneva PAROS

Nevertheless, despite initial expectations, weak enforcement and restricted legal binding of PAROS indicated the implications of unified national positions on definitional norms of space weapons and the scope of regulation. In particular, the dual-use nature of several space technologies suggested the intricacy of distinguishing peaceful satellite services from military capabilities; furthermore, a lack of consensus on the scope of regulation marked the complexity of the disclosure. As an attempt to overcome such concerns, the United Nations has established an Open-Ended Working Group to investigate exhaustive aspects of PAROS from 2025 to 2028, targeting to examine a combination of legally binding and non-binding measures. Still, the persistent strategic mistrust among member states of establishing universal and verifiable norms remains a significant challenge to impede arms race of space technologies.

UN General Assembly Resolution 77/41

The United Nations General Assembly urged member states to make a voluntary commitment of refraining from conducting destructive direct-ascent anti-satellite (ASAT) missile tests, which generate a significant amount of space debris, by adopting Resolution 77/41 on Destructive Direct-Ascent ASAT Missile Testing. Rather than comprehensively prohibiting the development or possession of space weapons, the resolution seeks to target the protection of the space environment and the alleviation of tensions among state governments by prioritizing restrictions on perilous space activities—including the destruction of orbiting satellites via ground-launched missiles. Yet, a lack of legal binding and restricted scope of counter-space capabilities—with co-orbital weapons, lasers, jamming, and cyberattacks excluded from the boundary on the resolution—marks its limitation in preventing an arms race between member states; consequently, the resolution is perceived as a mechanism to establish initial confidence-building measures aimed at emphasizing the necessity of long-term international normative frameworks.

Key Players

United States

A nation with the world's most advanced military and civil space capabilities, the United States extensively utilizes satellite communications, reconnaissance, missile warning technologies, and space systems for national security. In 2019, it established the US Space Force in order to manage space as an independent domain of military operations—a movement considered to raise concerns among rival nations that the US was expanding its military dominance in space. At the same time, the United States asserted the importance of impeding a future arms race of space technologies, actively supporting transnational measures, including the PAROS, that inhibit the utilization of space weapons and restrict risky actions; its position was exemplified by pledging to forgo testing ASAT missiles and urging other nations to follow suit in 2022. Thus, the United States is considered the world's most

Figure 5: U.S Space Force personnel conducting satellite operations

influential leading actor in developing space technologies and preventing a space arms race, while facing the critical challenge of balancing its own military space capabilities with international norms.

China and Russia

The Chinese and Russian governments have continuously expressed concerns that US military space capabilities would threaten the transnational armed strategic balance and asserted that the deployment of weapons should be strictly restricted by establishing an international treaty. The two member states jointly submitted a draft Treaty on the Prevention of the Placement of Weapons in the Outer Space and of the Threat or Use of Force Against Outer Space Objects (PPWT) to the Conference of Disarmament in 2008; and promoted the No First Placement initiative, suggesting that states pledge not to be the first to deploy weapons in space. Yet, the PPWT faces criticism for lacking sufficient or specific definition of space weapons and verification mechanisms, as well as for failing to effectively restrict ground-launched anti-satellite missiles. Given that both member states have generated massive amounts of space debris by destroying satellites—China's Fengyun-IC in 2007 and Russia's Cosmos-1408 in 2021—their position marks a key proponent of a comprehensive space arms management treaty while demonstrating consistent effort to develop and test their own anti-satellite and counter-space capabilities.

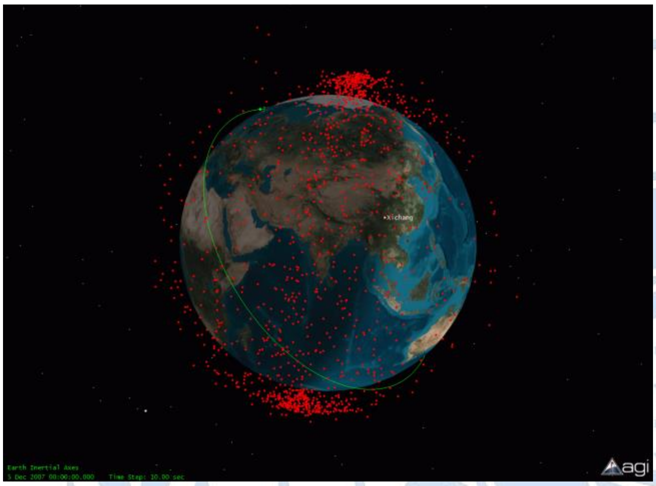

Figure 6: Fengyun-IC debris cloud NASA visualization in 2007 Chinese ASAT test

Possible Solutions

Strengthening transparency and confidence-building measures

To prevent an accelerating arms race in space technology, the international community should consider expanding existing Transparency and Confidence-building Measures (TCBMs) to reduce misunderstandings and mistrust. Member states should regularly disclose releasable information regarding their space policies and military space doctrines, while expanding the registration of space objects and designation of national points of contact. Furthermore, it is advisable to consult with associated nations—via diplomatic channels or existing Space Situational Awareness (SSA) systems—when conducting proximity maneuvers or hazardous tests that possess risks of deteriorating safety of neighboring satellites. UNOOSA could support registration of space objects, the management of national points of contact and sharing of non-military technical information, while the PAROS and the Conference on Disarmament (CD) could handle official disclosure regarding military norms. Through these efforts, risks of misjudgments regarding military expansion on space technologies could be mitigated as member states could strengthen diplomatic trust without disclosing secretive military information.

Expanding international commitments against destructive ASAT testing

The international community should consider expanding national commitments to refrain from conducting destructive direct-ascent anti-satellite (ASAT) missile tests, building upon the UN General Assembly Resolution 77/41. To this end, major space powers and nations with satellite launching capabilities should be encouraged to declare a voluntary moratorium on such tests, and participating states should regularly report to the UN on their relevant policies and the status of their commitment. The scope of commitments could be gradually broadened to include not only direct-ascent missiles but also any test involving the intentional in-orbit destruction of a satellite that generates long-lasting debris, further aiming at generating a more peaceful orbital environment. Through this approach, risks involved in ASAT missile testing could be restricted without requiring member states to abandon anti-satellite technologies.

Key Terms

Arms Race

A competitive process in which states continuously expand their military space capabilities in response to the actual or perceived capabilities of other states

Militarization of Outer Space

The use of space-based systems, including communication, navigation, reconnaissance, and early-warning satellites, to support military operations

Counter-space Capabilities

Technologies designed to disrupt, degrade, disable, or destroy another actor's space systems, including physical, electronic, cyber, and directed-energy capabilities.

Anti-Satellite Weapon (ASAT)

A weapon or system intended to damage, disable, or destroy a satellite through physical impact or non-physical interference.

Co-orbital ASAT system

A spacecraft placed into orbit that approaches another satellite and may inspect, interfere with, damage, or disable it.

Dual-use space technology

Space technology that can serve both peaceful civilian purposes and military objectives, making its intended use difficult for other states to determine.

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